

# 18.600: Lecture 18

## Uniform random variables

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Uniform random variable on  $[0, 1]$

Uniform random variable on  $[\alpha, \beta]$

Motivation and examples

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- ▶ Probability of any single point is zero.
- ▶ Define **cumulative distribution function**
$$F(a) = F_X(a) := P\{X < a\} = P\{X \leq a\} = \int_{-\infty}^a f(x)dx.$$

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- ▶ So  $\text{Var}[X] = E[X^2] - (E[X])^2 = 1/3 - 1/4 = 1/12$ .

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- ▶ What is the general moment  $E[X^k]$  for  $k \geq 0$ ?

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- ▶ Answer:  $\text{Var}[X] = \text{Var}[(\beta - \alpha)Y + \alpha] = \text{Var}[(\beta - \alpha)Y] = (\beta - \alpha)^2 \text{Var}[Y] = (\beta - \alpha)^2 / 12$ .

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- ▶ The way I defined it,  $p$  is uniform from the set  $\{0, 1/(n-1), 2/(n-1), \dots, (n-2)/(n-1), 1\}$ . When  $n$  is large, this is kind of like a uniform random variable on  $[0, 1]$ .

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- ▶ 2. The location of the first raindrop to land on a telephone wire stretched taut between two poles.
- ▶ 3. The amount of time you have to wait until the next subway train come (assuming trains come promptly every six minutes and you show up at kind of a random time).

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- ▶ 4. The amount of time you have to wait until the next subway train (without the parenthetical assumption above).

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- ▶ 6. The location of the ace of spades within a shuffled deck of 52 cards.